King Saud University
College of Computer and Information Sciences
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Semester-2 1447H

Eyecept Report

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CHAPTER.1

Introduction

1 Introduction

Security in high-value environments such as museums and jewelry exhibitions extends beyond merely monitoring physical spaces. It encompasses protecting valuable assets, preserving cultural heritage, and ensuring the safety of visitors. Since valuable items are often openly displayed and relatively accessible, these locations naturally become potential targets for theft attempts. Despite the widespread deployment of modern surveillance cameras, most existing surveillance systems primarily focus on recording incidents after they occur, rather than understanding behavioral patterns or enabling early intervention when risk indicators begin to emerge.

In practice, theft rarely occurs suddenly. It is typically preceded by subtle behavioral signals that may reflect emerging intent before an actual incident takes place. Although these behaviors may appear ordinary or insignificant at first glance, analyzing them within their temporal and environmental context may reveal a gradual escalation in risk levels.

In crowded environments, security personnel are required to monitor multiple screens simultaneously, creating continuous cognitive pressure and increasing the likelihood that early warning signals may be overlooked. As this operational burden accumulates, relying solely on human observation becomes insufficient for detecting risks at their earliest stages. In high-value settings, delayed detection may lead to financial losses and institutional reputational damage, making early identification of risk indicators critically important.

Furthermore, many current intelligent surveillance solutions remain reactive in nature, generating alerts only after a clear violation has already occurred. This limitation highlights the need for more advanced systems capable of supporting real-time behavioral analysis, reducing false alarms, and assisting security teams in prioritizing risks more effectively.

To address these challenges, this project proposes an intelligent surveillance system named Eyecept. The system is designed to support security teams in detecting early risk indicators before they escalate into actual threats. It aims to transform surveillance from a reactive process focused solely on post-incident documentation into a proactive analytical tool that enhances operational efficiency and informed decision-making. This report presents the identified problem, the proposed solution, the project objectives and scope, and an overview of the system architecture.

1.1 The Problem

Surveillance systems are widely used in high-value environments such as museums and jewelry exhibitions to monitor visitor movement and document events as they occur. While these systems provide essential support for observation and post-incident review, they remain limited in their ability to interpret subtle behavioral signals that may indicate emerging risk before an actual incident takes place.

In many real-world situations, security incidents do not happen suddenly. Instead, they are often preceded by minor behavioral cues that may appear ordinary or insignificant at first glance, especially in dynamic and crowded environments. For example, a visitor may repeatedly direct their attention toward a specific valuable item without physically interacting with it. Although such behavior may seem harmless when viewed in isolation, its repetition within a short period of time may reflect a developing risk pattern. However, current surveillance systems lack the analytical depth necessary to reliably distinguish between natural curiosity and behavior that may signal an elevated level of risk.

Moreover, most existing surveillance systems are primarily designed to document clear violations after they occur, reinforcing a reactive rather than preventive operational model. As a result, early behavioral indicators often go unnoticed, particularly when they are subtle, fragmented, or highly context-dependent. This creates a critical temporal gap between the emergence of the first warning signal and the moment effective intervention becomes possible.

Although some commercial solutions incorporate technologies such as motion tracking or pose estimation, significant limitations remain in their ability to interpret behavioral context, assess sustained attention patterns, and support proactive risk identification without generating excessive false alarms. Consequently, a clear operational gap persists in current surveillance practices, particularly in detecting subtle pre-incident behavioral indicators before they escalate into actual threats.

1.2 The Solution

In high-value environments such as museums and jewelry exhibitions, most surveillance systems focus on recording incidents after they happen, which limits the ability of security teams to respond early. Subtle behaviors that often appear before a theft attempt are usually overlooked, especially in crowded areas.

For example, a visitor may stand in front of a valuable item, keep their head and gaze directed toward it for an extended period, and slowly move their hand closer to the display. Since no clear violation has occurred, traditional systems do not flag this behavior, even though it may indicate a potential theft attempt.

The Eyecept system addresses this issue by focusing on a small set of reliable pre-theft indicators: hand movement, head orientation, and attention duration. These behaviors are analyzed locally using computer vision techniques, without relying on cloud processing. When unusual patterns are detected, the system sends a real-time alert to security staff, enabling early intervention while maintaining visitor privacy.

1.3 Product

In high-value environments like museums and jewelry exhibitions, most surveillance systems are designed to record what happens rather than prevent it. By the time an incident is flagged, it is often too late for security teams to intervene effectively. The problem is that suspicious behavior rarely starts with the theft itself — there are usually subtle signs that appear well before anything happens, and these early indicators tend to go unnoticed entirely

1.3.1 Product Vision

Product Vision: For a museum and jewelry exhibition security teams Who require early detection of suspicious behavior The Eyecept is a web-based intelligent surveillance system That detects pre-theft behaviors in real time Unlike traditional CCTV systems like Avigilon Video Analytics Our product analyzes subtle behaviors locally while preserving privacy

1.3.2 Product Roadmap

Figure 1- displays the roadmap for the Eyecept platform.

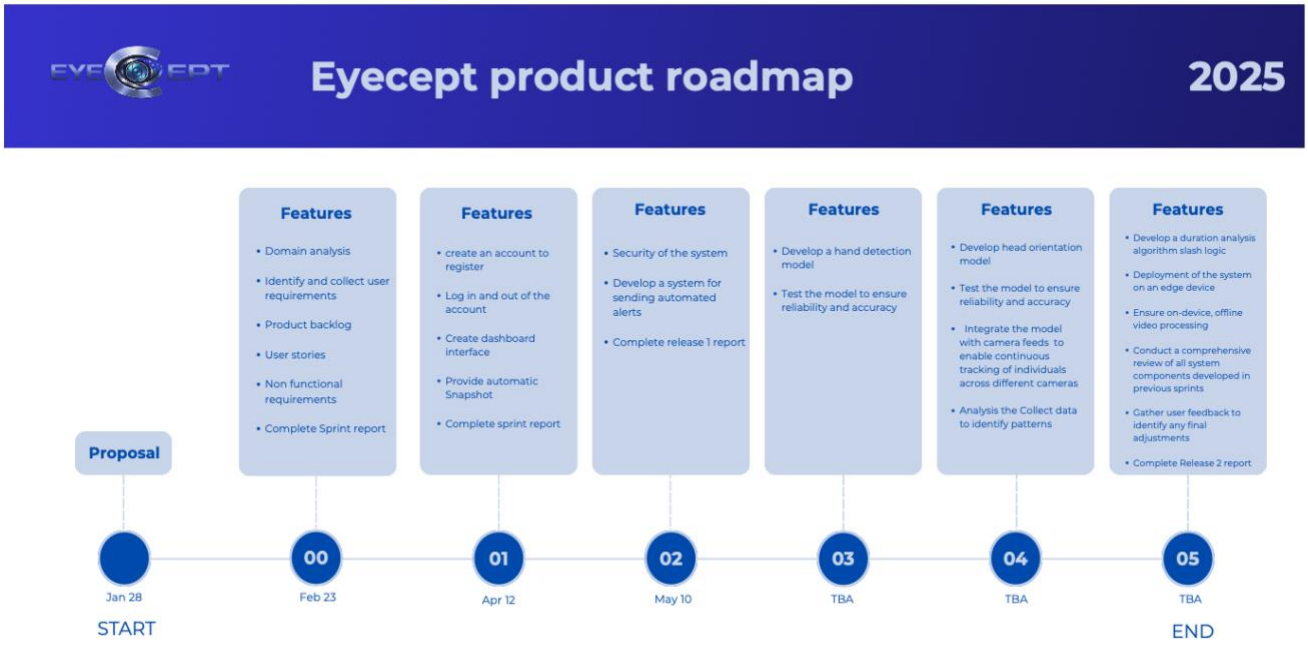


Figure 1: The Eyecept Platform Roadmap

This figure shows the Eyecept platform roadmap, outlining the project timeline from the proposal phase through multiple development sprints. It highlights key features, system development stages, testing, and final deployment.

1.3.3 Objectives

Product (customer focus-value):

Eyecept is a smart surveillance system that is used in high-value locations like museums and jewelry shows. The system facilitates early identification of suspicious behavior through examination of behavioral pointers in real time. It helps security guards by giving them instant alerts but all the processing of video is done locally to maintain privacy and minimize time-to-response.

The product objectives of Eyecept are as follows:

1. **Enable real-time hand movement detection:** Identify and detect hand movements around valuable items to detect suspicious activities before physical contact.
2. **Analyze head orientation and attention duration**: Track head pose and the length of time an individual focuses on a specific item to infer sustained interest that may indicate pre-theft intent.
3. **Detect protected-zone intrusions:** Establish sensitive spatial areas around exhibits of high value and raise alerts in case hands or bodies intrude into these spaces too soon.
4. **Provide instant suspicion notifications to security guards:** Send real-time notifications in case suspicious behavior patterns are identified to facilitate more efficient and quicker human reaction.
5. **Ensure on-device, offline video processing:** Perform all analysis locally on edge devices to preserve privacy, reduce latency, and eliminate dependence on cloud connectivity.
6. **Support multi-camera suspect tracking:** Keep constant surveillance of a target of interest from several camera perspectives within the facility.
7. **Documented proof in the form of an automatic snapshot or short incident video:** Record and hold timestamped image evidence on the local machine to be reviewed, reported, and analyzed after the fact.

- **Project (solution focus-plan):**

The Eyecept project will require the following steps to be completed:

1. **Requirements analysis and use-case definition:** Find out the security requirements in museums and jewels exhibitions, pre-theft behavior conditions, secure areas, and alarm states.
2. **Selection and preprocessing of online datasets:** It concerns the selection of appropriate publicly available surveillance databases, screening of the pertinent situations, and an initial processing of video data (frame extraction, labeling refinement, and quality correction) to fit the specific environment.
3. **Model selection and integration:** Pre-trained models of hand detection, head pose and person tracking need to be integrated and be trained on CCTV-like video.
4. **Development of behavior analysis logic:** Develop logic to analyze hand movement, head orientation ,and attention duration ,and use this information to determine early suspicious behavior.
5. **Implementation of zone and intrusion detection:** Design value-based areas of interest (ROIs) around valuable objects and identify the intrusion before physical contact.
6. **Evidence and alerting system:** Introduce real-time notifications to the security guards and automatically record snapshots or short clips with timestamps.

7. **Edge deployment and performance optimization:** Deploy the system on an edge device and optimize performance for offline, on-device processing.
8. **Monitoring dashboard development:** Develop a simple web-based dashboard to visualize alerts, events, and recorded incidents.
9. **Testing, evaluation and refinement:** Test the system with a variety of cases ,and camera views, determine the accuracy and latency, and make changes to the system.
10. **Documentation and final integration:** Final preparation and documentation of the system, and preparation for the final demonstration and evaluation.

- **Learning (student focus):**

1. **Computer vision on surveillance systems**: Practical expertise in integrating and deploying statistical models of computer vision like YOLO to detect objects, and hands, Hopenet to estimate the head pose, and ByteTrack to track people in actual CCTV applications.
2. **AI system design based on behavior:** Developing the ability to transform raw model outputs into useful behavioral attributes through integration of a series of indicators (hand movement, attention duration and spatial context) to identify the early signs of suspicious behavior.
3. **Edge AI and on-Device processing:** Practical experience deploying AI models on edge devices, optimizing performance for real-time, offline processing, and understanding the trade-offs between cloud-based and local computation.
4. **Multi-camera coordination and tracking: The** knowledge of how several cameras can team up and follow a target subject in various positions and still have the same identity throughout an area under surveillance.
5. **Alerts and notifications in real-time:** How to create and establish event-driven alerts and push notifications to alert the security staff immediately in case suspicious activity has been detected.
6. **Local event logging and storage of data:** Capability to achieve experience with the local storage of snapshots, incident clips, and alert metadata, as well as data flow between the processing of AI, storage, and the monitoring dashboard.
7. **Agile system integration and planning:** Training to break down a complex system into manageable modules, scheduling work into sprints, managing progress on a task management tool, and incorporating of multiple components into a working system.

1.3.4 Scope

The Eyecept project aims to enhance security in museums and jewelry exhibitions by detecting early suspicious behavior using video analysis. The system will process video locally from USB or IP cameras and provide real-time alerts to security personnel. It will initially support the English language and operate as an academic prototype in controlled environments. The project focuses on behavioral detection (head orientation, hand movement, and protected-zone intrusion) and does not include identity recognition or large-scale commercial deployment. The scope may be refined as development progresses.

1.3.5 Hardware/Software Tools and Cost

Table 1- displays the hardware/software tools and cost for the Eyecept platform.

Table 1:The Eyecept Platform Hardware/Software Tools and Cost Overview

Hardware Tools	
Name and Description	Cost *mention date it will be available*
Laptop for coding	No cost
USP or IP camera	250–500 SAR
Raspberry Pi 5 / Jetson Nano for edge processing	800–1500 SAR
External SSD storage	400-700 SAR
Stable network connection for alerts	200-600 SAR
Software Tools	
Name and Description	Cost
Word for documentation	No cost
Jupyter Notebook or google Collab	No cost
Python 3.10+	No cost
PyTorch	No cost
OpenCV	No cost
Ultralytics YOLO	No cost

ByteTrack or Norfair (tracking)	No cost
Hopenet (head pose estimation)	No cost
NumPy / Pandas	No cost
Roboflow for video processing	No cost
Average Total Cost	2475 SAR

1.4 Scrum Team

1.4.1 Skill Set Requirements

Table 2- displays the key skills requirement for the team of the Eyecept platform.

Table 2:The Eyecept Platform's Team Key Skill Requirements.

Technical Skill Required	What is the current level of the team (*beginner-intermediate- advanced*) for each skill? How will the gap be bridged? *(if necessary*) Learning plan
Python Programming Language	Intermediate
Computer Vision & Deep Learning	beginner
Video Processing and Image Analysis	beginner
Edge-Based AI Deployment	beginner
Database Design & Local Storage	Intermediate
Web-Based Monitoring Interface Development	Advanced

1.4.2 Learning

In the planning stage of the Eyecept project, the team developed a better ability to break down complicated real-world issues and express them into organized system elements. The team also learnt how to break down a massive intelligent surveillance system into manageable subdivisions such as detection, tracking, behavioral analysis, alert generation, and dashboard monitoring.

Another practicable experience that the team gained was the utilization of Jira to manage agile projects, including organizing the items of a product backlog, setting priority requirements, breaking tasks into manageable units, meeting minutes, and various project artefacts. As a result, the team was able to come up with a clear understanding of structured work workflows and collaborative project management. Besides this, the project enhanced the research and analysis skills of the team. The team has examined various scholarly articles and commercial systems, contrasted the available solutions, found research gaps, and used the same in streamlining and validating the proposed system design. In addition, the team also acquired a better understanding of the value of edge-based processing in privacy-sensitive settings and how architectural choices have the potential to affect system performance, latency, and data protection directly.

1.4.3 Roles and Responsibilities

Table 3- displays the roles and responsibilities for the team of the Eyecept platform

Table 3:The Eyecept Platform's Team Roles and Responsibilities Overview

Scrum Team	
Product Owner: (list name of the project supervisor)	Dr. Ameera Almasoud
Developers: (list student names)	Fatema AlHashem Joud Aljamaan Shahad Alotaibi Lana AlHumaidan
Scrum Master (SM): (list name of designated SM for the project)	Dr. Ameera Almasoud
Stakeholders: (list names of project stakeholders, include customer names if any)	Dr. Ameera Almasoud, Examination team

CHAPTER.2

BACKGROUND AND LITERATURE REVIEW

2 Background

2.1 Intelligent Surveillance Systems

Intelligent surveillance systems are advanced monitoring systems that go beyond traditional camera-based observation. Traditional surveillance systems mainly focus on recording video footage, which is later reviewed after an incident occurs. In contrast, intelligent surveillance systems use computational techniques to analyze video streams in real time and assist in detecting unusual or suspicious activities as they happen.

These systems aim to reduce the reliance on constant human monitoring, which can be challenging in crowded environments where security personnel are required to observe multiple screens simultaneously. By automatically analyzing visual data, intelligent surveillance systems help identify potential risks earlier, allowing faster and more effective responses.

In high-value environments such as museums and jewelry exhibitions, early detection is especially important. Suspicious behavior often appears before an actual incident occurs, and intelligent systems are designed to recognize these early indicators rather than only reacting after a violation has taken place.

2.2 Computer Vision in Surveillance

Computer vision is a branch of artificial intelligence that empowers machines to interpret and analyze visual information from images and video content. In surveillance systems, computer vision plays a pivotal role by providing automated understanding of what is happening within a monitored environment.

By processing video frames continuously, computer vision techniques can extract valuable information such as the presence of individuals, their movements, and their interactions with the environment, thereby reducing reliance on human observation. This capability enables real-time analysis, which is fundamental for detecting potential risks at the earliest possible stage.

In intelligent surveillance applications, computer vision is frequently utilized for tasks like object detection, tracking, and behavior analysis. These tasks are typically powered by modern deep learning models that can effectively handle complex visual patterns and dynamic environments with greater accuracy than traditional methods.

2.3 Object Detection and Person Detection

Object detection is a computer vision technique used to identify and locate specific objects within an image or video frame. In surveillance systems, it is commonly used to detect people and monitor their presence within a monitored area.

Detecting individuals is important because it allows the system to focus on human activity rather than irrelevant background elements. By identifying where a person is located, further analysis can be performed on their behavior, movement patterns, and interactions with objects in the environment.

Modern object detection approaches often rely on deep learning models such as **YOLO (You Only Look Once)**, which are designed to perform fast and accurate detection in real time. These models are widely used in surveillance systems due to their efficiency and ability to process continuous video streams.

In addition to person detection, detecting specific body parts such as hands provides more detailed behavioral insight. Hand movements, especially near valuable objects, may indicate potential interaction or intent. Frameworks such as **MediaPipe Hands** are commonly used to detect and track hand positions within video frames, enabling more precise analysis of movement patterns.

By combining person detection with hand detection, surveillance systems can move beyond simple monitoring and begin to interpret behavior, forming the foundation for higher-level analysis such as identifying suspicious activities.

2.4 Head Pose Estimation and Attention Analysis

Head pose estimation is a computer vision technique used to determine the orientation of a person's head, indicating the direction in which they are looking. In surveillance systems, this information helps in understanding a person's focus and attention within a monitored environment.

In high-value settings such as museums and jewelry exhibitions, prolonged attention toward a specific object may indicate unusual interest. While looking at an item is normal behavior, repeatedly focusing on the same object for an extended period may suggest a higher level of intent that requires attention.

Modern approaches to head pose estimation often rely on facial landmark detection models. Frameworks such as **MediaPipe Face Mesh** can be used to detect key facial points and estimate head orientation in real time. These tools enable systems to analyze attention patterns more accurately within dynamic environments.

By analyzing both the direction and duration of a person's gaze, surveillance systems can move beyond simple observation and begin to identify patterns that may indicate suspicious behavior.

2.5 Behavioral Analysis in Surveillance

Behavioral analysis in surveillance refers to the process of interpreting human actions and patterns over time in order to identify unusual or potentially risky behavior. Unlike traditional surveillance systems that focus on detecting clear violations, behavioral analysis aims to recognize subtle indicators that may appear before an incident occurs.

In real-world environments, suspicious activity often develops gradually rather than occurring suddenly. For example, repeated attention toward a specific object, combined with slow or unusual movements, may indicate emerging intent. These behaviors may seem normal individually, but when analyzed together over time, they can reveal meaningful patterns.

Behavioral analysis combines multiple inputs such as object detection, hand movement, and attention tracking to form a more complete understanding of what is happening in the scene. This approach allows surveillance systems to shift from reactive monitoring to proactive detection.

However, one of the key challenges in behavioral analysis is reducing false positives. False positives occur when normal behavior is incorrectly identified as suspicious. In surveillance systems, this can reduce trust in alerts and increase workload for security personnel. Therefore, maintaining a balance between sensitivity and accuracy is essential to ensure reliable system performance.

2.6Region of Interest (ROI)

A Region of Interest (ROI) is a specific area within an image or video frame that is selected for focused analysis. Instead of processing the entire scene equally, surveillance systems can prioritize certain areas that are more critical or sensitive.

In high-value environments, ROIs are typically defined around valuable objects or restricted zones. By concentrating analysis within these areas, the system can more effectively monitor interactions that are most relevant to security.

When movement or activity occurs within the ROI, it is analyzed more closely to determine whether it represents normal behavior or a potential risk. This targeted approach improves efficiency and reduces unnecessary processing of irrelevant regions.

ROI-based monitoring also helps in improving detection accuracy, as it allows the system to focus on meaningful interactions rather than background noise. As a result, it plays an important role in supporting real-time decision-making in intelligent surveillance systems.

2.7 Edge Computing

Edge computing refers to processing data locally on a device rather than sending it to a remote cloud server. In surveillance systems, this approach allows video data to be analyzed directly at or near the source, such as on a local machine or edge device.

One of the main advantages of edge computing is reduced latency. Since data does not need to travel to a remote server for processing, results can be generated in real time, which is essential for timely detection of suspicious behavior. This enables faster response and more effective decision-making in security scenarios.

Another important benefit is improved privacy. By processing video data locally, sensitive visual information does not need to be transmitted or stored externally, reducing potential privacy risks. This is particularly important in environments such as museums and exhibitions, where maintaining visitor privacy is a priority.

Overall, edge computing supports efficient, real-time, and privacy-aware surveillance, making it a suitable approach for intelligent monitoring systems.

2.8 Surveillance Cameras and Video Streaming

Surveillance cameras are the main hardware component in monitoring systems, as they provide the visual input needed to observe and analyze activity within an environment. In surveillance applications, cameras are placed in strategic locations to cover important areas and reduce blind spots, allowing security teams to maintain awareness of what is happening across the monitored space.

Depending on the environment and system setup, cameras may be connected in different ways. Some systems use USB cameras connected directly to a local device, while others use IP cameras that transmit video over a network. In many real-world surveillance environments, multiple cameras are used together to provide wider coverage and observe events from different angles.

Video streaming refers to the continuous transmission of video frames from these cameras to the processing system. In real-time surveillance, this stream is analyzed frame by frame so that ongoing activity can be monitored as it happens. This makes video streaming essential for systems that aim to detect suspicious behavior without delay.

The effectiveness of surveillance analysis depends heavily on the quality and stability of the video stream. Factors such as camera resolution, frame rate, viewing angle, and network reliability can all affect the system's ability to detect events accurately. For this reason, surveillance cameras and video streaming are not only technical components, but also a fundamental part of the overall reliability of intelligent monitoring systems.

2.9 Event Detection and Alert Systems

Event detection in surveillance systems refers to identifying activities or behavior patterns that may indicate a potential security concern. These events may include unusual movement, repeated attention toward a specific object, or activity within a protected area.

When such an event is detected, the system generates an alert to notify the appropriate monitoring side. Alerts may contain supporting information such as the time of occurrence, the source camera, and captured visual evidence. This helps connect the detected event to a clear security context.

In intelligent surveillance systems, alerts are often organized into different levels according to the seriousness of the detected behavior. Surveillance systems commonly use multi-level alert mechanisms, where different levels correspond to increasing degrees of risk and may trigger different forms of monitoring, notification, or escalation. Lower-level alerts may indicate behavior that is worth monitoring further but does not yet require direct intervention, while higher-level alerts represent stronger indicators of risk and therefore trigger a more immediate response. In this way, alert levels do not simply rank events by importance, but also support different stages of monitoring and escalation.

An effective alert system should therefore do more than detect activity. It should also communicate the right level of concern in a clear way, so that security personnel can respond appropriately without being overwhelmed by unnecessary notifications. Alerts may be delivered through monitoring dashboards or other communication channels, depending on the level and type of response required.

2.10 Local Data Storage and Database Systems

Surveillance systems often require storing data such as detected events, captured images, and system logs for later review and analysis. Local data storage refers to saving this information directly on a device within the system, rather than transmitting it to external cloud services.

One of the main advantages of local data storage is enhanced privacy. Since sensitive visual data is processed and stored within the system, it is not exposed to external networks, reducing the risk of unauthorized access. This is particularly important in environments such as museums and exhibitions, where privacy and data protection are critical.

Local storage also supports continuous operation without relying on internet connectivity. The system can function independently, ensuring that event detection, alert generation, and data recording continue even in offline conditions. This makes it suitable for real-time surveillance scenarios where reliability is essential.

Lightweight database systems such as SQLite are commonly used in local storage environments. These databases allow efficient management of structured data, including user information, alert records, timestamps, and associated media paths.

While different storage approaches exist depending on system requirements, local data storage remains a practical solution for applications that prioritize privacy, low latency, and on-device processing. It plays a key role in supporting real-time, secure, and self-contained surveillance systems.

3 Literature Review

The rapid development of Artificial Intelligence and Computer Vision technologies has significantly transformed modern video surveillance systems. Traditional surveillance solutions mainly focus on monitoring and recording events after they occur, which limits their ability to prevent security incidents before they happen. Recently, research has shifted toward intelligent video analytics systems capable of detecting suspicious behaviors in real time and supporting proactive security decision-making [1].

In high-value environments such as museums and jewelry exhibitions, early detection of suspicious behavior is critical. Studies in behavioral analysis suggest that subtle indicators such as prolonged attention toward valuable objects, repeated head orientation, and abnormal hand movements near secured items can indicate potential malicious intent before an actual theft attempt occurs. Real-world security incidents in high-value environments demonstrate how quickly theft events can occur and highlight the need for intelligent preventive monitoring systems [2]. However, many existing surveillance solutions are designed to detect obvious events such as motion, intrusion, or object removal rather than subtle pre-incident behavioral patterns.

Several intelligent surveillance and video analytics solutions currently exist in the global market. For example, Veesion provides AI-based video analytics designed to detect shoplifting gestures and suspicious hand movements in retail environments. Similarly, Vaidio is an AI video analytics platform widely used in smart cities and enterprise surveillance, offering features such as abnormal behavior detection, intrusion detection, and object recognition. Additionally, companies such as Hikvision, Dahua, Bosch, and Axis Communications provide enterprise-level surveillance systems

focusing on motion detection, object tracking, and intrusion alerts as part of general security monitoring solutions.

Furthermore, biometric and behavior analytics companies such as 3DiVi focus primarily on identity recognition technologies, including face recognition and body tracking, which are mainly used for access control and identification rather than behavioral prediction. Regional providers such as ACIX Middle East offer integrated physical security solutions focusing on operational monitoring, crowd movement tracking, and post-incident analysis.

Despite the strong capabilities of these systems, most of them follow a reactive security model that focuses on detecting events after they occur. In addition, many solutions rely heavily on cloud-based processing, which may introduce latency and raise privacy concerns, especially in sensitive environments such as museums and exhibitions. Edge computing has emerged as an important alternative approach, enabling video data to be processed locally on the device, which reduces latency, improves response time, and enhances data privacy [3].

Recent research trends highlight the importance of edge computing in surveillance systems. Edge-based processing enables video data to be processed locally on the device, reducing latency, improving response time, and enhancing data privacy. This approach is particularly important in environments where data confidentiality is critical [3].

Based on the identified limitations in existing systems, the Eyecept system is designed to provide a proactive behavioral security solution that focuses on detecting pre-theft behaviors using multi-factor behavioral analysis. The system combines real-time hand movement detection, head orientation analysis, and attention duration monitoring to identify suspicious behavioral patterns around high-value objects.

In addition, Eyecept supports protected zone monitoring using Region of Interest (ROI) analysis and generates instant alert notifications when suspicious behavior is detected. The system also supports multi-camera suspect tracking, automatic short-clip recording upon detection of suspicious activity and provides a centralized monitoring dashboard for real-time security monitoring and decision-making.

Eyecept is also designed using on-device edge processing, which enhances privacy protection and reduces dependence on cloud infrastructure. This design makes the system suitable for deployment in privacy-sensitive environments such as museums, jewelry exhibitions, and high-security facilities [3].

Moreover, Eyecept is particularly suitable for deployment in regions such as the Middle East and Saudi Arabia, where there is an increasing demand for intelligent security solutions that ensure privacy preservation while maintaining high detection accuracy and rapid response capabilities.

Overall, the literature review demonstrates the need for proactive behavioral security systems that go beyond traditional surveillance monitoring. The analysis of existing solutions highlights the research gap in detecting subtle pre-incident behaviors. Eyecept aims to address this gap by providing an intelligent, privacy-preserving, and real-time behavioral surveillance solution specifically designed for high-value asset protection environments.

3.1 Competitive Product Analysis

1-Veesion

Figure 2-Veesion Logo

The logo shown above belongs to **Veesion**, a French company that provides an AI-based video analytics system designed to detect shoplifting gestures and suspicious hand movements in retail environments. The system focuses on recognizing theft-related gestures and sending real-time alerts to security personnel.

Unlike Eyecept, which is deployed in museums and jewelry exhibitions and targets pre-theft behavior using subtle indicators such as prolonged attention and gradual hand movement, Veesion mainly detects theft during or immediately before the act. Moreover, Veesion relies on cloud-based processing, while Eyecept operates entirely on edge devices to ensure privacy preservation and on-site real-time response. [4]

2-Vaidio

Figure 3-Vaidio Logo

The logo shown above belongs to **Vaidio**, an AI-powered video analytics platform developed by IronYun and widely used for security surveillance, smart cities, and transportation monitoring. It provides features such as abnormal behavior detection, intrusion detection, and object and person recognition.

In contrast to Eyecept, which focuses specifically on detecting subtle pre-theft behaviors in high-value environments such as museums and jewelry exhibitions, Vaidio is a general-purpose surveillance system that analyzes overall movement and anomalies in the scene rather than fine-grained attention and hand movement patterns. Furthermore, Eyecept is designed for privacy-preserving edge deployment, while Vaidio often relies on centralized or cloud-based architectures. [5]

3-DiVi

Figure 4-3DiVi Logo

The logo above belongs to **3DiVi**, a computer vision and biometric technology company that develops systems for face recognition, body tracking, and behavior analysis. Its solutions are used in security and access control applications to identify individuals and analyze their movements.

Unlike Eyecept, which aims to predict pre-theft behavior using subtle cues such as head orientation and attention duration toward valuable objects, 3DiVi mainly focuses on identity recognition and general behavior analytics. Additionally, Eyecept is designed explicitly for privacy-sensitive environments and operates locally on edge devices, reducing privacy risks associated with biometric identification systems. [6]

4- **Hikvision**

Figure 5-HikiVision Logo

The logo above belongs to **Hikvision**, a Chinese company specializing in large-scale CCTV implementations, focusing on person and vehicle detection, motion analytics, and intrusion alerts as part of comprehensive security surveillance solutions.

Eyecept is unique in its setting and purpose, as it is deployed in museums and jewelry exhibitions, to prevent pre-theft behaviors. Instead of relying on general movement in the

scene, it utilizes subtle indicators (e.g., prolonged attention and hand movements in secured areas). Since Eyecept operates entirely locally on edge devices, it can react on-site and maintain data confidentiality. [7]

5- **Dahua Technology**

Figure 6-Dahua Technology Logo

The logo above belongs to **Dahua Technology**, a Chinese company that offers surveillance systems primarily designed to monitor visible activity through motion and person detection, and as well as broad behavior analytics, sometimes with cloud storage capabilities.

Eyecept shifts the analysis from scene-level activity to object-level risk by computing hand movement within item-associated ROIs and directly activating real-time audio cues and instant notifications. It also has the advantage of being lightweight and edge-based, making it easier to deploy while enhancing privacy in sensitive settings. [8]

6-**Bosch**

Figure 7-Bosch Logo

The logo above belongs to **Bosch**, a German company that offers sophisticated video analytics capable of generating alerts when motion or suspicious activity is detected. The data can be stored either on premises or in the cloud, with a focus on actions occurring within the environment.

Eyecept, in contrast, was created specifically for museums and jewelry exhibitions and is oriented toward recognizing hand movement before or during a theft attempt. Unlike Bosch systems, Eyecept implements a threat-level-based alerting model, where alert responses increase in strength, ranging from monitoring to dashboard alerts or direct security warnings. [9]

7-Axis Communications

Figure 8-Axis Communications Logo

The logo shown above belongs **to Axis Communications**, a Swedish company and a global leader in network video surveillance solutions. It provides high-quality IP cameras and video analytics used in airports, smart cities, and large enterprises.

Axis systems mainly focus on monitoring and recording events after they occur, with limited capabilities in detecting subtle pre-theft behaviors. Additionally, Axis solutions are generally considered high-cost compared to lightweight edge-based systems such as Eyecept. [10]

8-**Hanwha Vision**

Figure 9-Hanwha Vision Logo

The logo shown above belongs to **Hanwha Vision**, a South Korean company specializing in AI-powered surveillance solutions. It offers smart cameras and video analytics for motion detection and object classification in commercial and public facilities.

Despite incorporating AI technologies, its solutions mainly focus on general surveillance tasks and do not specifically analyze detailed pre-theft behavioral indicators such as head orientation or attention duration toward valuable objects. [11]

9-**ACIX Middle East**

The logo shown above belongs to **ACIX Middle East**, a regional provider of integrated physical security and surveillance solutions, offering video analytics as part of comprehensive security systems.

Figure 10-ACIX Middle East Logo

Its analytics primarily focus on detecting observable events within monitored environments, such as unauthorized access, zone intrusions, crowd movements, and general activity tracking. These solutions are mainly intended to support operational security and post-incident investigation rather than interpreting subtle behavioral cues. [12]

3.2 Comparison for competitor Systems

The competition analysis table that would be compared with the Eyecept would be the comparative analysis of Eyecept with some of the most successful surveillance and video analytics systems. The main functional and architectural features that can be taken to the high value environment such as museums and jewelry shows are juxtaposed. All these are hand gesture recognition, focus and head location, security zone monitoring, real-time alarm, screen edges, multi-camera tracking, edge detection and dashboard, automatic evidence capture and dashboard detection.

Table 4:Comparison of competitors

Features / competitors	Real-time hand movement detection	Head orientation & attention duration analysis	Protected-zone (ROI) intrusion alert	Instant suspicion notifications	On-device, offline edge processing	Multi-camera suspect tracking	Automatic snapshot / short-clip recording	Monitoring dashboard	Solution Types
Veesion	✓			✓			✓	✓	Retail video Analytics
Vaidio			✓	✓	✓	✓	✓	✓	AI vision platform
3Divi				✓	✓	✓	✓	✓	Biometric + CV analytics
Hikvision			✓	✓	✓	✓	✓	✓	Enterprise CCTV + analytics
Dahua Technology			✓	✓	✓	✓	✓	✓	Enterprise CCTV + analytics
Bosch			✓	✓	✓	✓	✓	✓	Video analytics
Axis Communication			✓	✓	✓	✓	✓	✓	Network video
Hanwha Vision			✓	✓		✓	✓	✓	AI survellance
Acix middle east			✓	✓		✓	✓	✓	Security solutions provider
Eyecept	✓	✓	✓	✓	✓	✓	✓	✓	Web monitoring

As indicated in the comparison in **Table 4**, the competitive advantages are those unique to the Eyecept system when compared to a number of well-known video analytics systems and intelligent surveillance systems in the world like behavioral analysis systems, biometric analytics systems, and the traditional enterprise surveillance systems. In contrast to the comparison, the fact that the Eyecept can be used to facilitate proactive security activities in high value environments like museums and jewelry shows depicts that the capability can be employed to prevent security incidents that would otherwise take a long way before security event would happen. Most of the available surveillance systems have powerful features like the motion detection, object tracking and recording of occurrences but are usually configured to give reactive response to the surveillance or analyze a situation which has already occurred. Eyecept, on the other hand, has pro-active security model, which is established on a multi-factor behavioral analysis. The system has real time hand movement recognition, head orientation recognition, and head attention time recognition so as to pick the tiniest behavioral clues of the threat of theft that is not always detectable by the conventional surveillance solutions. Furthermore, Eyecept offers the possibility to track the secure zones via the analysis of the Region of Interest (ROI) and an immediate notification was made in case of the deviant or suspicious behavior. This will allow the security personnel to respond quickly hence averting an attack before security breakage. Moreover, Eyecept will be compatible with on-device edge processing that can improve the confidentiality of the information, does not require the utilization of the cloud infrastructure, and limits the system latency. The system also allows the multi-camera tracking of the culprits where upon the detection of suspicious activities, automatic short clip recording is done and also allows central control monitoring dashboard that can be used by the security team to make real-time decisions relating to security matters. This type of architecture renders Eyecept very well adapted in such a setup with a high privacy sensitivity rate as well as in places like the Middle East and Saudi Arabia where an increasing demand of smart security systems exists in ensuring the safety of privacy and at the same time, high detection rates and speed. This leads to the fact that Eyecept has a good potential of being practically implemented and working successfully in a real-life environment with high security.

CHAPTER.4

System Requirements

4 System Requirements

This section introduces the Eyecept system by identifying its primary users and outlining their roles and experience within security monitoring environments. It then defines the users' needs based on insights gathered from interviews and questionnaires. Furthermore, the system architecture is presented to provide a clear overview of the main components and explain how they function together. Finally, a Use Case Diagram is included to illustrate how users interact with the system in practical scenarios.

4.1 System Users

Security Operators describe people who normally work in a surveillance and security monitoring area like an exhibition, or a commercial place, and they are usually aged 23 years and above. They usually have a diploma or a bachelors degree and experience in the running of monitoring systems and responding to security incidents. They have medium technical skills and are not necessarily advanced in artificial intelligence knowledge; however, they are accustomed to working with web-based platforms and control dashboards.

Administrators are experts who compile and arrange the system of the organization. They are usually above 27 years old and have at least a bachelor degree in Information Technology, Computer Science or any other similar course. They possess average to high level technical skills, and are experienced in system administration, networking and database administration.

Field Security officers are working employees whose duty is to physically react to alerts and deal with possible threats on the ground. They are more likely to be above 21 years of age, with a high school education level, and they are professionally trained in threat management and emergency response. They have rudimentary technical expertise, enough to deal with mobile alerts and communication devices.

4.2 Requirements Elicitation and Analysis

In order to have a clear definition of what would be needed in the Eyecept system, we did a requirements elicitation to gain a clearer insight into the actual problems that professionals working in high-sensitivity surveillance face. The aim was to understand the nature of monitoring as it is currently practiced, where existing systems pose challenges, and what enhancements can possibly be made to enable quicker and more effective security decision-making.

There were two methods of data collection. To begin with, semi-structured interviews were carried out in person with security professionals operating in exhibitions and financial monitoring environments. These interviews gave us the opportunity to have structured questions and at the same time provide us with the flexibility to have the participants elaborate on their experiences, daily challenges, and concerns in greater detail. Second, a structured questionnaire was distributed via Google Forms, and 31 replies were gathered. The questionnaire covered workload monitoring, expectations of the reaction time, system usability, the impact of false alarms, and the openness of AI-assisted support. An open-ended question was also included to gather additional qualitative feedback.

The participants had between 9 and 18 years of experience in surveillance-related roles. Their responses reflect the reality of environments where continuous attention is required, and even small delays in identifying suspicious behavior can result in serious consequences.

The insights gathered from both the interviews and the questionnaire directly supported the refinement of the system's functional and non-functional requirements.

(Appendix A :Interviews for full interview transcripts and Appendix B: Questionnaires for detailed questionnaire results.)

4.2.1 Interview Analysis:

1. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage rather than during live monitoring?

Participants explained that suspicious behavior is sometimes identified only after reviewing recorded footage rather than during live monitoring. In addition, access to recordings may require administrative approval, which can delay verification and slow down response actions.

In high-value environments, this workflow increases operational risk. These findings support the need for a proactive surveillance system capable of detecting suspicious behavior in real time without relying on post-incident review procedures.

2. Do you believe that relying solely on human observation makes it difficult to consistently detect small or slow movements in busy exhibition environments?

Interviewees highlighted that relying exclusively on human attention makes it difficult to consistently detect subtle or slow suspicious behaviors, particularly in crowded exhibition environments or during peak hours. They also described the mental demands of monitoring multiple camera feeds for extended periods, noting that sustained concentration increases the likelihood of fatigue and oversight.

This reinforces the need for intelligent behavioral analysis to support operators and reduce the cognitive burden associated with continuous monitoring.

3. How difficult is it to track a person of interest when they move between different camera views or between multiple exhibition areas?

Tracking a person of interest across multiple camera views was described as time-consuming and mentally demanding. Participants indicated that manually following individuals between areas or camera feeds increases workload and reduces situational clarity, especially in busy environments.

This finding strengthens the case for automated multi-camera tracking to improve continuity, reduce operator effort, and support faster decision-making.

4. What factors would make you trust and rely on an AI-based surveillance system in your daily work?

Participants emphasized that trust depends primarily on accuracy, reliability, and the clarity of alerts. They stated that frequent false alarms reduce confidence in the system, whereas alerts supported by visual evidence enhance trust and facilitate faster validation.

These insights highlight that reliability, low false-positive rates, and transparent alert explanations are essential for user acceptance of AI-based systems.

5. During peak hours, how manageable is it for you to monitor all camera screens while maintaining the same level of attention continuously?

Interviewers explained that maintaining consistent attention during peak hours is challenging due to the number of screens and the density of visitor movement. Extended monitoring periods increase mental fatigue, which raises the probability of missing important events.

This supports the necessity of automated assistance to maintain consistent surveillance performance during high-demand periods.

6. When a suspicious situation is identified, what are the usual steps taken by the security team, and where do delays typically occur?

Participants described a multi-step process involving footage review, administrative approval, and coordination with field officers. Delays most commonly occur during verification and approval stages.

Such procedural delays create operational gaps in high-value environments, further emphasizing the importance of real-time detection and immediate alerting mechanisms.

7. What type of alert would be most effective for you during work (visual dashboard alert, sound notification, mobile notification, or a combination of these)?

Participants expressed a preference for immediate and noticeable alerts, ideally combining visual and sound notifications. They also stressed that alerts should include contextual information explaining the reason for activation.

This indicates that multi-modal alerting supported by contextual explanation enhances operational responsiveness.

2.10.1.1 Questionnaire Analysis:

Q1. How many hours per shift do you spend monitoring cameras?

A substantial proportion of respondents reported extended monitoring durations per shift. Specifically, 45.2% reported monitoring for 5–8 hours, 22.6% for 2–4 hours, 16.1% for less than 2 hours, and 16.1% for more than 8 hours.

These findings demonstrate that surveillance tasks require prolonged attention, reinforcing the presence of sustained cognitive load in daily operations.

Q2. During monitoring, how high is your mental workload or focus level required?

The results indicate that monitoring demands significant mental effort. Specifically, 43.8% reported high workload, 40.6% reported medium workload, and 15.6% reported very high workload.

This confirms that surveillance roles involve considerable cognitive pressure, increasing the risk of fatigue-related errors.

Q3. How often do you feel that important events might be missed during live monitoring?

More than half of respondents (51.6%) reported that important events are sometimes missed, while 29% indicated that this occurs often.

These findings highlight the limitations of manual monitoring and strengthen the need for automated behavioral detection support.

Q4. How quickly do you usually need to react when suspicious behavior is noticed?

The data reveal strong expectations for rapid response. Specifically, 43.8% require action within seconds, and 37.5% expect response within one minute.

This emphasizes the operational necessity of low-latency detection and immediate alert delivery.

Q5. How clear and easy to use is your current surveillance system interface?

While 51.6% rated their current system interface as neutral in clarity, 22.6% described it as difficult to use.

This suggests that current surveillance interfaces may not fully support rapid and confident decision-making under pressure.

Q6. How important is it for you to receive alerts with visual evidence (snapshot or short clip)?

An overwhelming majority considered visual evidence essential. Specifically, 61.3% rated it as very important, and 32.3% rated it as important.

This strongly validates the requirement for evidence-supported alerts in the proposed system.

Q7. How much do false alarms affect your work efficiency?

False alarms were identified as a significant concern. Specifically, 38.7% reported a moderate impact, while 22.6% reported a significant impact on their efficiency.

These findings indicate that minimizing false positives is critical to maintaining operational efficiency and trust in the system.

Q8. How useful would it be if the system prioritized the most risky behavior automatically?

A strong majority expressed support for automatic risk prioritization. Specifically, 58.1% rated it as very useful, and 41.9% rated it as useful.

This demonstrates clear demand for intelligent threat-level classification.

Q9. How comfortable are you with AI assisting you in monitoring tasks?

Acceptance of AI assistance was generally high. Specifically, 61.3% reported feeling comfortable, and 25.8% reported feeling very comfortable.

This indicates broad readiness among professionals to adopt AI-supported surveillance tools.

Q10. How important is system reliability (working without crashes or delays) during monitoring?

System reliability was rated as very important by 74.2% of respondents.

This confirms that stability, uptime, and consistent performance are non-negotiable requirements in high-sensitivity surveillance environments.

4.2.2 Consolidated Key Findings

Based on both qualitative and quantitative findings, the following conclusions were identified:

- Surveillance roles involve sustained cognitive workload and high mental pressure.
- Reactive monitoring and procedural delays create operational vulnerabilities.
- Human-only observation is insufficient for detecting subtle or slow suspicious behaviors.
- Real-time alerts supported by contextual visual evidence are operationally critical.
- False alarms negatively affect efficiency and reduce trust in the system.
- Automatic risk prioritization is strongly supported by users.

- AI-assisted monitoring is widely accepted among professionals.
- System reliability and low latency are essential performance requirements.

These findings provide a structured foundation for defining and refining the functional and non-functional requirements in the Product Backlog, ensuring that the proposed Eyecept system directly addresses real operational challenges in high-value surveillance environments.

4.3 User Interactions

The use case diagram illustrates the main interactions between external actors and the Eyecept system, focusing on key functions such as live monitoring, alert handling, protected zone (ROI) definition, and user management. The Security Operator monitors live video feeds and reviews alerts generated from suspicious activity within defined protected zones, while the Admin manages user accounts and overall system access.

The Edge Device processes video streams from cameras by applying detection rules based on presence within the ROI, duration, and repeated appearances. When a potential risk is detected, a snapshot is captured and stored. Based on these conditions, the system determines the threat level: for Level 1, the system continues monitoring and may signal other cameras; for Level 2, the alert is sent to the dashboard; and for Level 3, the alert is sent to the dashboard and an SMS notification is triggered through the SMS Gateway to notify the Field Security Officer.

The transmitted alert includes relevant information such as snapshots, timestamps, dates, and assigned threat levels.

Alert handling allows users to view alerts with their details and optionally mark them as false positives (dismissed). Admin operations include adding, updating, and removing users, while both the Admin and Security Operator can reset their passwords when needed.

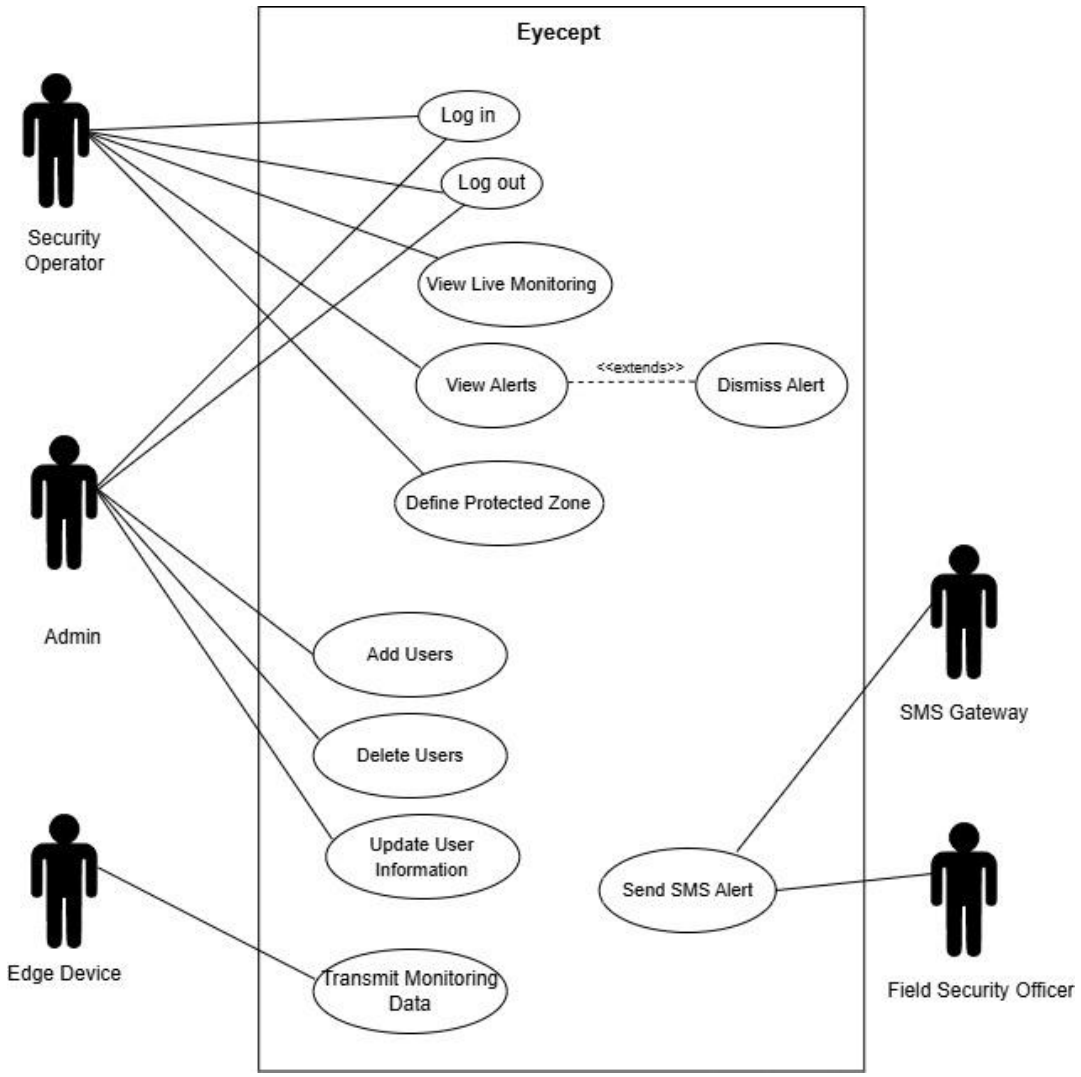


Figure 11:Use Case Diagram

Figure 12:System Architecture

4.4 Product Backlog

It is a part which shows the Product Backlog of the Eyecept system. It contains a set of prioritized features and user stories, which are required to deploy the system in small increments. The items in the backlog are referred to the project objectives as well as system requirements and are organized depending on the development priority. In addition to that, the Definition of Ready is provided to have a clear definition of every item in the backlog, which will be technically viable, and ready to be rolled out to the implementation in a sprint.

4.4.1 Definition of Ready

Table 5: Definition of Ready

Defentions of Ready	
✓	Details are sufficiently understood
✓	Acceptance criteria are clear and testable
✓	Estimated and small enough to be completed during sprint
✓	Dependencies are identified; no blocking dependencies exist
✓	Performance criteria, if any, are defined and testable

2.10.2 Product Backlog Table

Table 6 displays the Product Backlog table for the Eyecept, and it contains the user stories for functional and non-functional requirements.

Table 6:PBI Table

PBI (user story)	Size (Story points)	Type (Feature, defect, technical work, knowledge acquisition)	Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted.
Functional Requirements			
As a security operator, I want to receive real-time alerts when suspicious behavior is detected so that I can take preventive action.	7	Feature	- If suspicious behavior is detected, then a real-time alert must be sent to security staff.
As a security operator, I want to receive a notification alert with a threat level so that I can understand the level of risk.	8	Feature	- If the same person is detected three or more times and attention duration exceeds 15 seconds, then a notification alert must be generated including the threat level.
As a security operator, I want a snapshot to be automatically captured when a threat alert is generated so that the event is documented.	5	Feature	- If an alert is generated, then a snapshot must be captured and stored in the event logs including threat level, date, and time.

As a security operator, I want a short incident clip to be automatically recorded when a threat alert is generated so that I can review the suspicious behavior in detail.	5	Feature	- If an alert is generated, then a short incident clip must be recorded. - The clip must be linked to the same event record as the alert.
As a security operator, I want to view the monitoring dashboard so that I can monitor detected events.	5	Feature	- If the user accesses the dashboard, then live alerts must be visible.
As a security operator, I want to view incident details so that I can understand detected suspicious behavior.	6	Feature	- If the user selects an alert, then the system must display the incident details.

As a security operator, I want to access event logs through the dashboard so that I can review detected incidents.	5	Feature	- If an event is logged, then it must be accessible through the dashboard interface.
As a security operator, I want to dismiss an alert if it does not require intervention so that false positives are reduced.	5	Feature	- If the security determines no intervention is required, then the alert must be dismissible.
As a field security officer, I want to receive an SMS notification immediately when a person's hand enters a predefined ROI so that I can respond immediately.	8	Feature	- If a person's hand enters a predefined Region of Interest (ROI), then the system must detect the intrusion. - If ROI intrusion is detected, then an immediate SMS notification must be sent to the field security officer.

As a security operator, I want to create a new account so that I can access the system.	5	Feature	- If the user provides valid required information, then the account must be created successfully. - If required information is missing, then the system must reject the registration request.
As a security operator, I want to log in to the system so that I can access the monitoring dashboard.	3	Feature	- If valid login credentials are entered, then the system must grant access to the dashboard. - If invalid credentials are entered, then access must be denied.
As an admin, I want to add new user accounts so that authorized staff can access the system.	4	Feature	- If the admin enters valid user information, then the account must be created successfully. - If required information is missing, then the system must reject the creation request.
As an admin, I want to delete user accounts so that unauthorized users cannot access the system.	4	Feature	- If the admin deletes a user account, then that user must not be able to log in. - If a deleted user attempts to log in, then access must be denied.

As a security operator, I want to log in to the system so that I can access the monitoring dashboard.	3	Feature	- If valid login credentials are entered, then the system must grant access to the dashboard. - If invalid credentials are entered, then access must be denied.
As a security operator, I want to log out of the system so that I can securely end my session.	3	Feature	- If the user selects the log out option, then the session must be terminated.
Non-Functional Requirements			
As a security operator, **I want** suspicious behavior alerts to appear on the dashboard within 3 seconds of detection **so that** I can act immediately	5	Non-Functional Requirement (NFR)	- As a security operator, if suspicious behavior is detected, then the alert must appear on the dashboard within 3 seconds from detection under normal operational load.
As a security operator, **I want** the system to calculate attention duration **so that** suspicious behavior can be detected.	7	Non-Functional Requirement (NFR)	- If a person remains focused for 7–12 seconds, then the activity must be flagged as suspicious. - If the activity is flagged as suspicious, then the system must instruct other cameras to begin monitoring the same individual.
As a security operator, **I want** the system to automatically detect camera disconnections and attempt reconnection within 30 seconds, **so that** monitoring continuity is maintained	5	Non-Functional Requirement (NFR)	- As a security operator, if a camera feed is disconnected, then the system must detect the disconnection and initiate a reconnection attempt within 30 seconds.
As an admin, **I want** video processing and behavior analysis to run locally with an average inference time of less than 200 milliseconds per frame, **so that** real-time	8	Non-Functional	-As an admin, if video frames are processed on the edge device, then the average inference time must not exceed 200 milliseconds per frame under normal operational load.

monitoring is maintained without noticeable delay.		Requirement (NFR)	
As a security operator, **I want** the monitoring dashboard to be available at least 95% of operational hours, **so that** security monitoring is not interrupted.	3	Non-Functional Requirement (NFR)	-As a security operator, if I access the monitoring dashboard during operational hours, then it should be available at least 95% of the time
As a security operator, **I want** all alerts and recorded snapshots/clips to be stored without any data loss **so that** no critical security evidence is missed	5	Non-Functional Requirement (NFR)	-As a security operator, if an alert or a snapshot/clip is generated, then it must be stored with a timestamp and remain accessible without data loss during normal system operation.
As an admin, **I want** the system to support monitoring of at least 4 cameras simultaneously and operate independently of camera brand**, so that** the system can scale to larger exhibition areas without hardware limitations	8	Non-Functional Requirement (NFR)	-As an admin, if I connect up to 4 cameras that support RTSP or ONVIF protocols, then the system must process all feeds simultaneously while maintaining the defined alert latency threshold.
As an admin, **I want** recorded snapshots and incident clips to preserve visitor privacy through anonymization techniques, **so that** individuals cannot be personally identified from stored evidence.	8	Non-Functional Requirement (NFR)	- As an admin, if a snapshot or incident clip is stored, then identifiable facial features must be anonymized, and the data must remain stored locally without transmission to external cloud servers.
Enabler Stories			
As a developer, **I need** to integrate the detection, tracking, and behavior analysis components into a unified processing pipeline **so that**	8	Enabler	-As a developer, if a video frame is captured, then the system must process it through detection, tracking, and behavioral analysis sequentially without runtime errors.

suspicious behavior can be analyzed in real-time.			
As a developer, **I need** to integrate multiple camera streams(IP/USB) into the system, **so that** real-time video frames can be processed by the AI pipeline.	5	Enabler	-As a developer, if a camera is connected, then the system must initialize the stream and capture frames successfully.
As a developer, **I need** to implement model loading and runtime inference, **so that** the trained AI model can process frames during system operation.	5	Enabler	-As a developer, if the system starts, then the trained model weights must load successfully and perform inference on incoming frames.
As a developer, **I need** to connect the back-end processing engine with the monitoring dashboard **so that** the alert and event logs are displayed to security personnel.	5	Enabler	-As a developer, if suspicious behavior is detected, then an alert must be transmitted from the back-end and displayed on the dashboard.
As a developer, **I need to** prepare and train the YOLO detection model using collected prefetch data sets, **so that** detection accuracy improves for suspicious actions	8	Enabler	-As a developer, if the dataset is prepared and used for training, then updated model weights must be generated and validated for improved detection performance.
As a developer, **I need** to integrate the trained models onto an edge device, **so that** processing runs locally without cloud dependency	8	Enabler	As a developer, if the model is deployed on the selected edge device, then video processing must run locally, and no video data shall be transmitted to external cloud services.

As a developer, **I need** to integrate a tracking algorithm with the detection model **so that** individuals can be tracked across frames.	5	Enabler	As a developer, if the same person appears in consecutive frames, then the system must maintain the same tracking ID under normal movement conditions.
As a developer, **I need** to implement risk scoring logic based on behavioral indicators **so that** threats of different levels can be classified.	5	Enabler	As a developer, if suspicious behavioral indicators are detected, then the system must calculate a risk score and classify the threat level
As a developer, **I need** to validate system stability under multi-camera load conditions, **so that** the system's deployment readiness can be ensured	5	Enabler	-As a developer, if 4 cameras streams are active, then the system must remain stable, and performance logs must be recorded.
As a developer, I need to integrate a local SQLite database into the edge-based system so that alerts, snapshots, and behavioral data can be stored securely on-device without relying on cloud connectivity.	5	Enabler	- As a developer, if the system starts for the first time, the SQLite database is created successfully. - As a developer, if a new alert is generated, it is stored in the database without errors. - As a developer, if stored data is requested, it is

			retrieved and displayed correctly. - As a developer, if the system is restarted, previously stored data remains available. - As a developer, if the system is offline, the database continues to function normally.

CHAPTER.5

System Design

5 System Design

2.11 Architectural Diagram

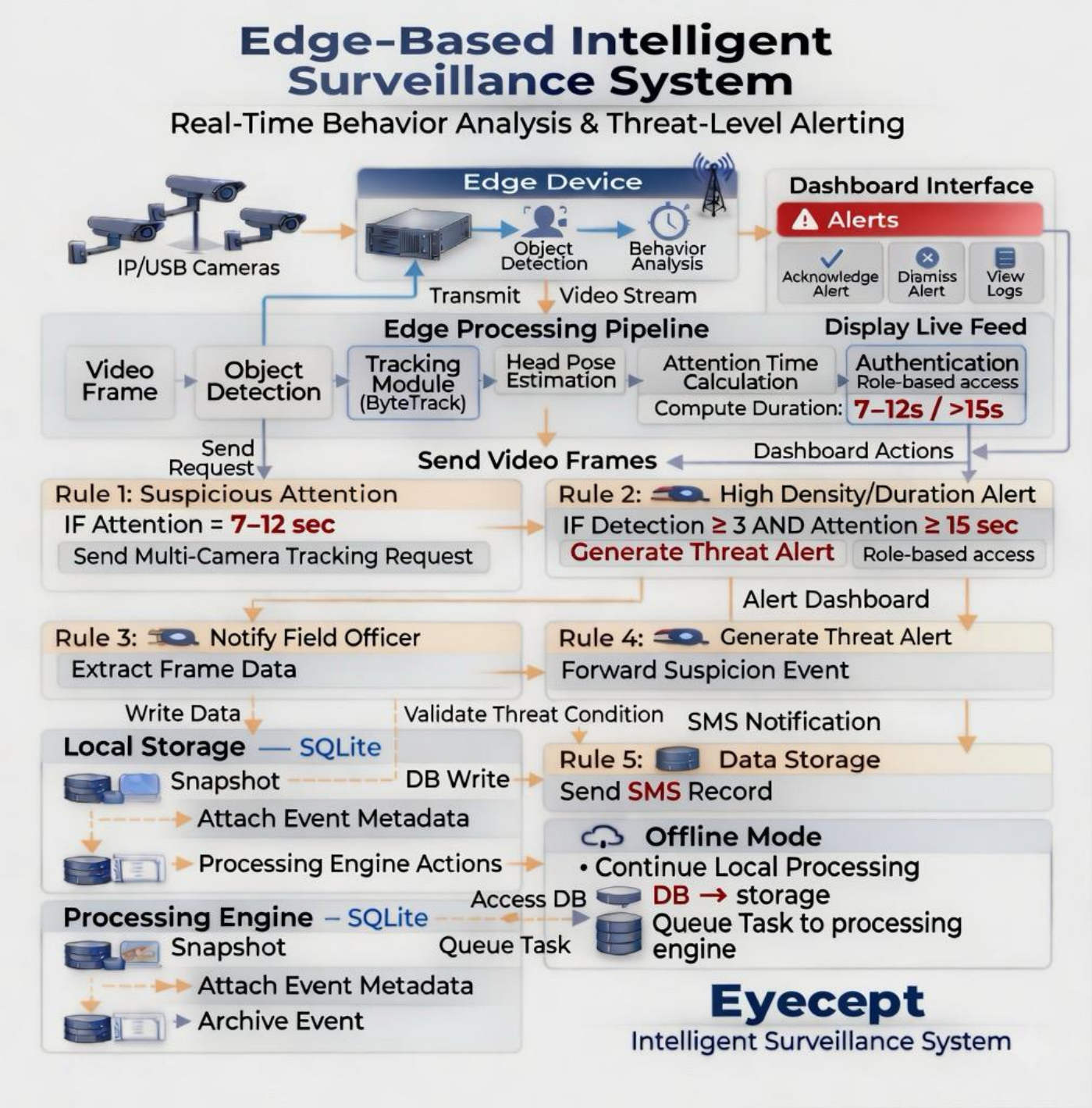
The Eyecept system follows a layered, client–server architecture designed to detect suspicious behavior in real time while preserving privacy. The system is organized into several major components: cameras, an edge processing device, a local SQLite database, a web-based monitoring dashboard, and an alert notification subsystem.

Live video from USB or IP cameras serves as the system input and is streamed to the edge device for real-time processing. The edge device performs person detection, movement tracking, and attention analysis to identify potential suspicious behavior. When suspicious patterns are detected, the system generates outputs such as alerts, snapshots stored in the local database, and notifications displayed on the monitoring dashboard or sent via SMS to field officers.

The system is divided into three security levels: Level 1 – system administrator responsible for configuration and database management; Level 2 – security operations officer monitoring the cameras and dashboard; and Level 3 – field security officer responding on-site to high-priority alerts.

These components interact as follows: cameras stream video to the edge device, which analyzes the video frames and records events in the SQLite database, and the dashboard retrieves this information to display alerts and monitoring information to authorized users

The layered design separates data capture, processing, storage, and user interaction, making the system easier to maintain, extend, and deploy in privacy-sensitive environments.


Edge-Based Intelligent Surveillance System
Real-Time Behavior Analysis & Threat-Level Alerting
IP/USB Cameras
Edge Device
Object Detection
Behavior Analysis
Transmit
Video Stream
Dashboard Interface
Alerts
Acknowledge Alert
Dismiss Alert
View Logs
Edge Processing Pipeline
Display Live Feed
Video Frame
Object Detection
Tracking Module (ByteTrack)
Head Pose Estimation
Attention Time Calculation
Authentication
Role-based access
Compute Duration: 7–12s / >15s
Send Request
Send Video Frames
Dashboard Actions
Rule 1: Suspicious Attention
IF Attention = 7–12 sec
Send Multi-Camera Tracking Request
Rule 2: High Density/Duration Alert
IF Detection ≥ 3 AND Attention ≥ 15 sec
Generate Threat Alert
Role-based access
Alert Dashboard
Rule 3: Notify Field Officer
Extract Frame Data
Rule 4: Generate Threat Alert
Forward Suspicion Event
Write Data
Validate Threat Condition
SMS Notification
Local Storage — SQLite
Snapshot
DB Write
Attach Event Metadata
Processing Engine Actions
Rule 5: Data Storage
Send SMS Record
Offline Mode
• Continue Local Processing
Access DB
DB → storage
Queue Task to processing engine
Processing Engine – SQLite
Snapshot
Queue Task
Attach Event Metadata
Archive Event
Eyecept
Intelligent Surveillance System

1.2 Class Diagram /DFD

Figure 13 displays the Class Diagram of the Eyecept system. There are three types of users in the system: Admin, Security Operator, and Field Security Officer. The Admin is in charge of user accounts, the Security Operator watches the dashboard, sees alerts, and sets up the ROI, and the Field Security Officer gets SMS alerts for important alerts.

Cameras stream video to the Edge Device, which processes the feed, detects suspicious behavior, captures snapshots, and creates alerts. These alerts are displayed on the dashboard, where they can be reviewed and marked as false positives, and in critical cases, the SMS Gateway sends notifications to the Field Security Officer.

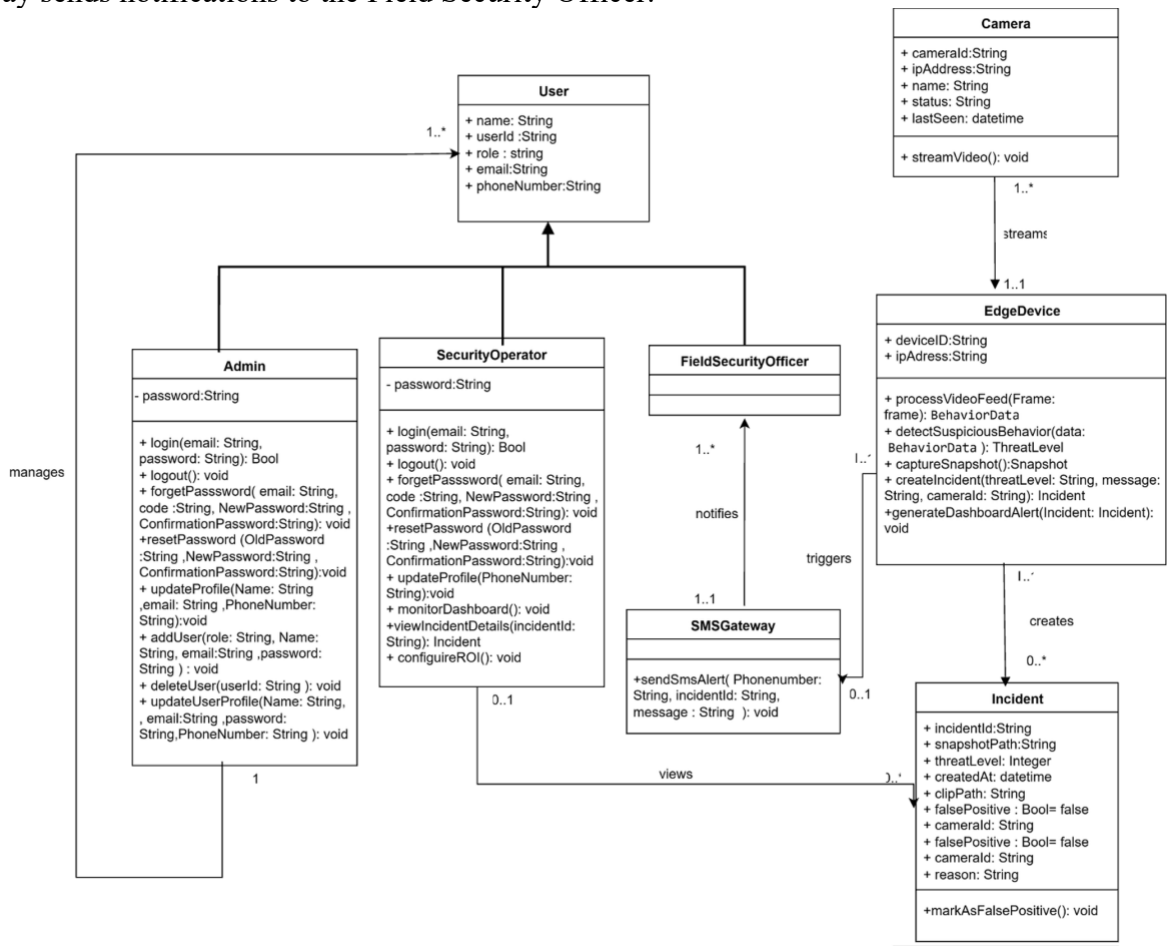


Figure 13:Class Diagram

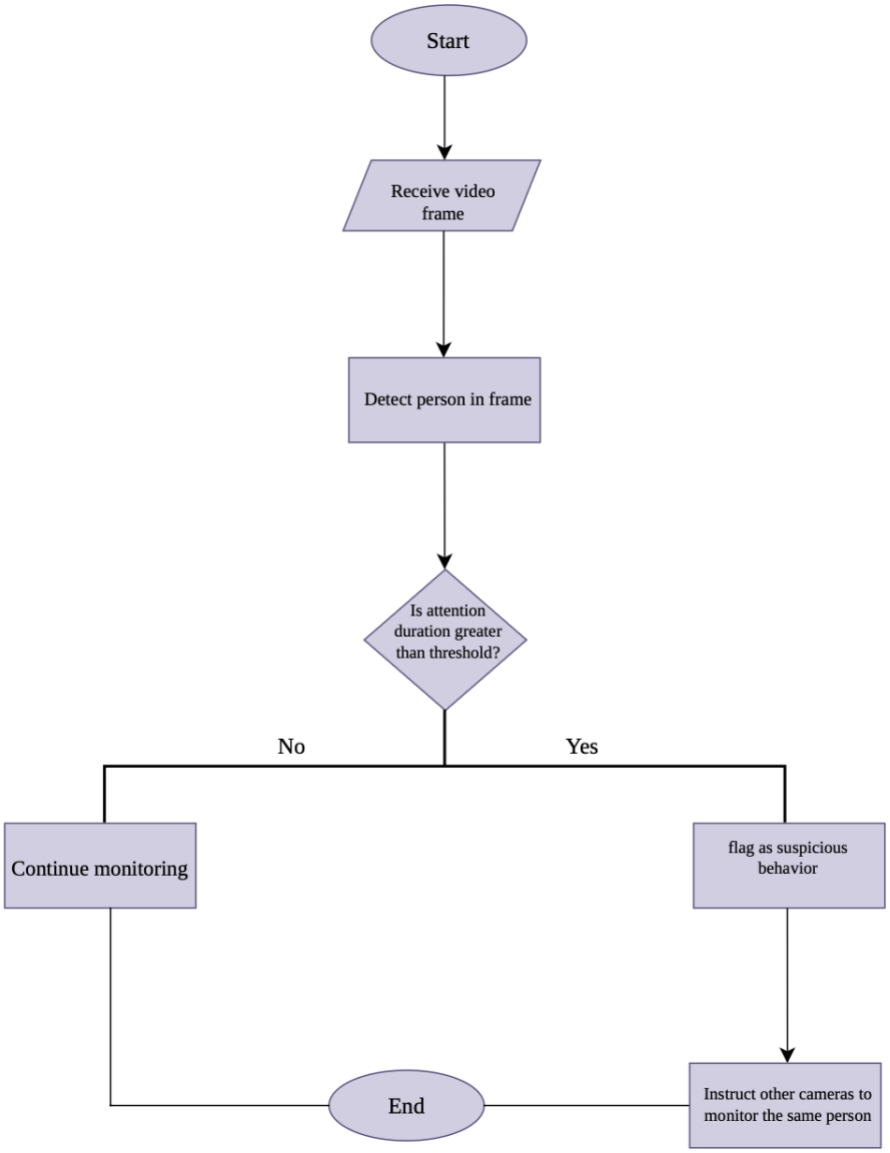
2.12 Component Level Design

This section explains how the main components of the proposed system are implemented at a detailed level. It focuses on illustrating the internal logic and operational flow of the system through structured representations such as flowcharts and step-by-step processes.

For clarity, the system is divided into several key components that represent the core functionalities of the solution, including behavior analysis, threat assessment and alert generation, and region-based intrusion detection. Each component is described individually to demonstrate how the system processes data, makes decisions, and produces appropriate responses in different scenarios.

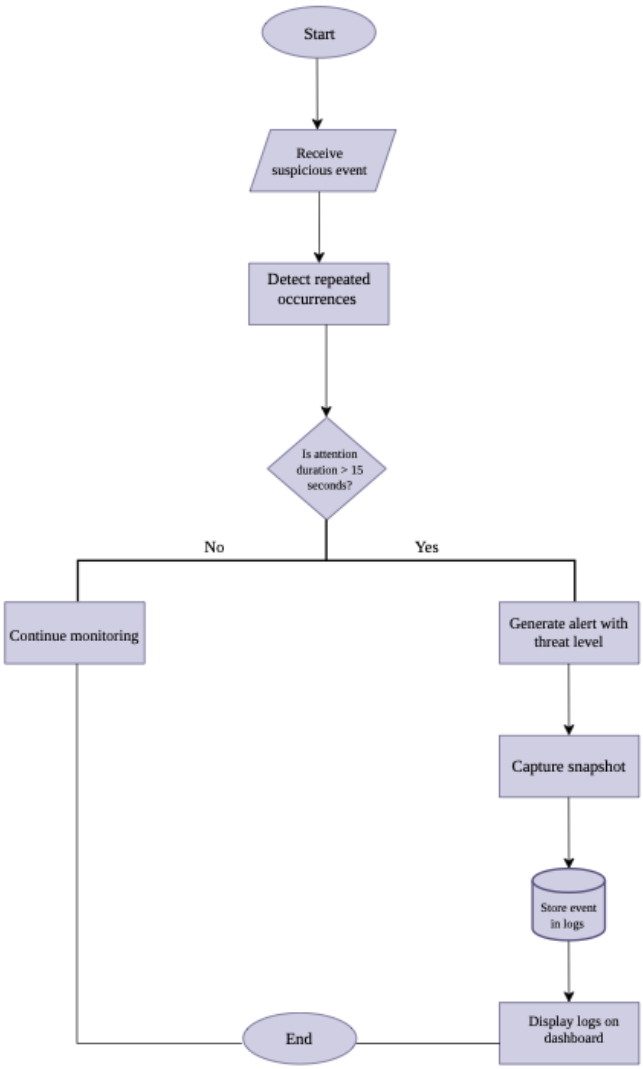
2.12.1 Suspicious Behavior Detection

This section presents the main processes involved in detecting suspicious behavior based on attention duration. The system continuously analyzes video frames to identify individuals and evaluate their focus duration. When a person maintains attention within a defined threshold, the activity is classified as suspicious, and further monitoring actions are triggered.



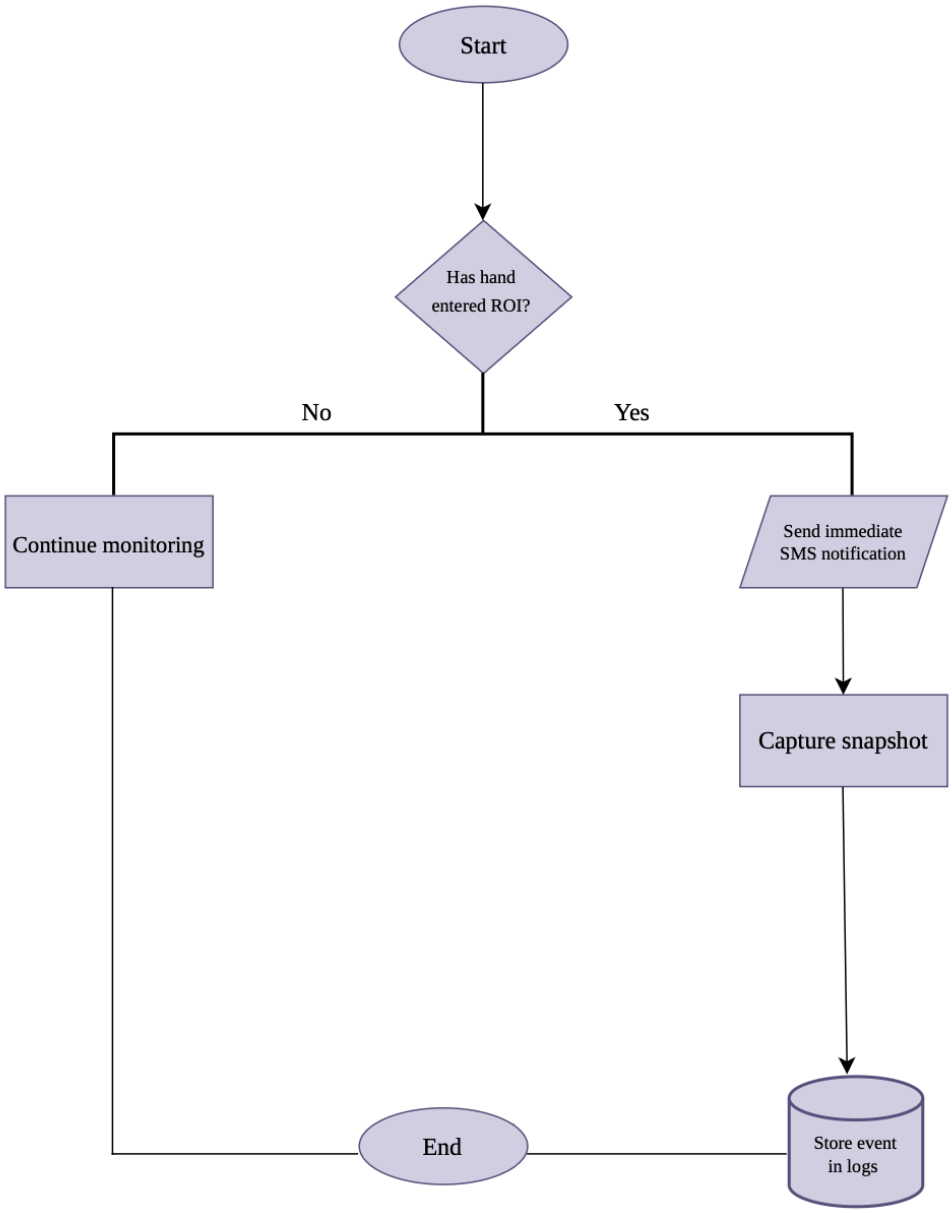
2.12.2 Threat Assessment and Alert Generation

This section describes the process of evaluating suspicious behavior to determine whether it represents a higher-level threat. The system analyzes repeated occurrences of suspicious activity along with extended attention duration to assess the severity of the situation and decide whether escalation is required.



2.12.3 Immediate Threat Detection and SMS Notification

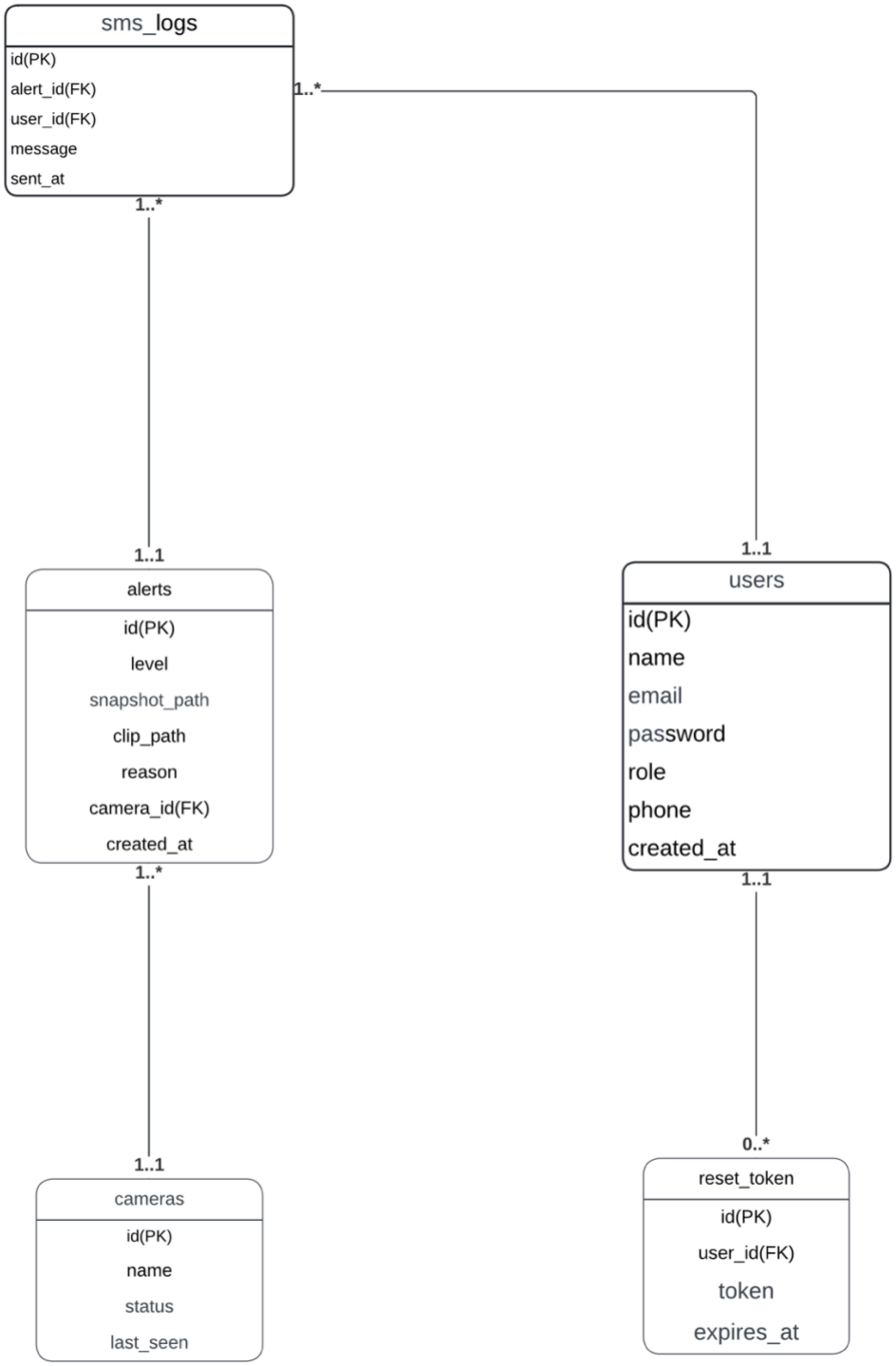
This section explains the mechanism for detecting immediate threats based on spatial intrusion. Unlike previous levels, this process does not rely on attention duration but instead focuses on identifying critical actions such as entering a predefined Region of Interest (ROI), which requires instant response.



2.13 Data design

2.13.1 Data models

This section describes the structure and organization of the data used in the proposed system. The data model is designed to support efficient storage and management of information generated from video analysis and security monitoring. The system data is organized into related entities such as users, cameras, edge devices, incidents, and alert notifications, with defined relationships that reflect how data is connected and maintained. The following subsections present the ER diagram, followed by the relational schema and data dictionary to provide a detailed representation of the database structure.


sms_logs
id(PK)
alert_id(FK)
user_id(FK)
message
sent_at
1..*
1..*
1..1
1..1
alerts
id(PK)
level
snapshot_path
clip_path
reason
camera_id(FK)
created_at
users
id(PK)
name
email
password
role
phone
created_at
1..*
1..1
1..1
0..*
cameras
id(PK)
name
status
last_seen
reset_token
id(PK)
user_id(FK)
token
expires_at

2.13.2 Relational schema

USERS (
id PK,
name,
email,
password,
role,
phone,
created_at
)

CAMERAS (
id PK,
name,
status,
last_seen
)

ALERTS (
id PK,
level,
snapshot_path,
clip_path,
reason,
camera_id FK ,
created_at
)

```
SMS_LOGS (
    id PK,
    alert_id FK,
    user_id FK,
    message,
    sent_at
)

RESET_TOKEN(
    id PK,
    user_id FK,
    token,
    expires_at
)
```

2.13.3 Data Dictionary

This table shows all entities and their attributes in the system, including their data types and descriptions.

Table 7: Data Dictionary

Entity name	Attribute	Description	Data Type	Length	Null	PK
User	id	Unique identifier for each user	INTEGER	3	No	Yes
	name	User full name	VARCHAR	50	No	No
	email	User email address	VARCHAR	100	No	No
	password	Encrypted password	TEXT	—	No	No
	role	User role (admin/operator/etc)	VARCHAR	17	No	No
	phone	User phone number	VARCHAR	10	Yes	No
	created_at	Account creation timestamp	TIMESTAMP	—	No	No
Cameras	id	Unique camera identifier	INTEGER	—	No	Yes
	name	Camera name	VARCHAR	20	No	No
	status	Camera status	VARCHAR	7	No	No
	last_seen	Last active time	TIMESTAMP	—	Yes	No
Alerts	id	Unique alert identifier	INTEGER	—	No	Yes
	level	Threat level	VARCHAR	10	No	No
	snapshot_path	Path to snapshot image	TEXT	—	Yes	No
	clip_path	Path to video clip	TEXT	—	Yes	No
	reason	Reason for alert	TEXT	—	Yes	No
	camera_id	Related camera ID	INTEGER	—	No	No
	created_at	Alert creation time	TIMESTAMP	—	No	No
Sms_logs	id	Unique SMS log identifier	INTEGER	—	No	Yes
	alert_id	Related alert ID	INTEGER	—	No	No
	user_id	Receiver user ID	INTEGER	—	No	No
	message	SMS content	TEXT	—	No	No
	sent_at	Time SMS was sent	TIMESTAMP	—	No	No

Reset_token	id	Unique token identifier	INTEGER	—	No	Yes
	user_id	Related user ID	INTEGER	—	No	No
	token	Password reset token	VARCHAR	6	No	No
	expires_at	Token expiration time	TIMESTAMP	—	No	No

2.13.4 Data collection and preparation

Data Collection

The system collects its data from two main sources, each serving a specific role in supporting the overall analysis process.

1. Real-Time Data

Video streams are captured directly from cameras using the OpenCV library. This setup reflects a real-world surveillance environment, where the system continuously monitors activities as they occur. As events unfold, relevant behavioral indicators are detected and processed immediately, allowing the system to respond with minimal delay.

2. External Data Sources

In some cases, external datasets may be utilized to enhance the system's performance. This is particularly beneficial during the fine-tuning stage, where pre-trained models are adjusted to better recognize subtle behavioral patterns or adapt to environments that differ from their original training conditions.

To support further processing, recorded video is also divided into individual frames, which are then used in later stages of analysis and preparation.

Data Preparation

Before being fed into the system, the data undergoes a preparation stage to ensure it is suitable for analysis. This stage involves several steps, including:

- Extracting frames from video recordings

- Resizing images to meet model input requirements

- Filtering out unclear or irrelevant frames

- Manually defining Regions of Interest (ROI) to highlight sensitive areas

- Organizing the data in a way that supports behavioral analysis

- Formatting the data to ensure compatibility with detection models

When necessary, labeled data can be used during fine-tuning to further improve model performance within the target environment.

Once the system is operational, it continuously extracts behavioral features in real time based on model outputs. These may include:

- Detecting when a hand enters a defined ROI

- Recognizing repeated appearances of the same individual

- Measuring how long a person focuses on a specific object or area

It is important to note that suspicious behavior is not determined by a single model alone. Instead, it is inferred by combining multiple signals generated by different models working together. This integrated approach helps produce results that are more accurate, reliable, and context-aware.

Tools and Algorithms

The system relies on a set of well-established tools and pre-trained models to perform its tasks effectively. These include:

- OpenCV — used for video processing, frame extraction, and handling live streams

- YOLOv8 (pre-trained on the COCO dataset) — used for real-time human detection

- MediaPipe Hands — used for tracking hand movements

- MediaPipe Face Mesh — used for estimating head orientation

Rather than building models from scratch, the system leverages pre-trained models and adapts them when necessary through fine-tuning. This approach allows the models to better suit high-value surveillance environments.

In addition, **SQLite** is used as the database system. Since it operates locally within the system, it enhances privacy and security by eliminating reliance on external servers. It is also lightweight and integrates easily with the application.

2.14 Interface design

The Eyecept system provides a web-based interface designed to support real-time monitoring and efficient interaction for different users, including administrators, security operators, and field security personnel.

Login Interface

The system includes a secure login page that allows authorized users to access the platform by entering their username and password. This ensures that only permitted users can interact with the system.

Admin Registration Interface

A dedicated interface is provided for administrators to register new users. Through this interface, the admin can add security operators by entering their required information. The system validates the input before creating the account, ensuring secure and controlled access.

Dashboard Interface

The main dashboard serves as the central interface for security operators. It displays real-time alerts generated by the system in a clear and organized layout. Each alert includes key information such as threat level, timestamp, camera ID, and a snapshot image, allowing operators to quickly assess situations.

Alert Interface

The system supports two types of alert notifications. SMS alerts are sent to field security personnel (security guards) to ensure immediate response on-site, while alert notifications are displayed on the dashboard interface for security operators to enable continuous monitoring and analysis.

Event Log Interface

The interface also includes an event log section where users can review previously detected incidents. Each record contains detailed information along with a snapshot image, supporting analysis and decision-making.

CHAPTER.6

System Implementation

7 System Implementation

The Eyecept system integrates computer vision techniques with web technologies to enable real-time behavioral analysis and alert generation. The system is built using Pyth and utilizes OpenCV for video processing, while detection is performed through specialized models where Person Detection is handled using YOLOv8 (COCO) to identify individuals in each frame, Hand Detection is performed using MediaPipe Hands to accurately track hand movements and interactions, and Head Orientation is analyzed using MediaPipe Face Mesh to estimate the direction of the user's head and attention.

After detection, ByteTrack is used to track individuals across frames, and the system analyzes behavior such as hand proximity to objects, head direction, and duration of attention to evaluate potential risk. When suspicious behavior exceeds predefined thresholds, the system generates an alert and captures a snapshot of the event. All detected events are stored locally in an SQLite database, including the snapshot image, timestamp, camera ID, threat level, and detection reason, ensuring user privacy by keeping all data on the local device without relying on cloud services.

CHAPTER.7

System Testing

8 System Testing

2.15 Experimental Results

2.16 User Acceptance Testing

User Acceptance Testing was conducted to evaluate whether the Eyecept system meets the needs of its intended users, particularly security operators working in surveillance environments. The testing process involved a group of participants with hands-on experience in surveillance and security monitoring environments, who were asked to interact with the system interface and evaluate both its usability and overall functionality. Participants were required to complete several tasks, including logging into the system, accessing the monitoring dashboard, viewing detected alerts, reviewing event logs, and interpreting snapshot evidence. The evaluation focused on key aspects such as ease of use, speed of alert notifications, clarity of visual evidence, and overall system reliability. The feedback collected indicated that the system interface was generally easy to understand and navigate, and that the presence of visual snapshot evidence significantly improved user confidence when validating alerts. Participants also noted that the combination of visual alerts and supporting evidence helped reduce uncertainty while evaluating suspicious situations. Overall, the results suggest that the Eyecept system provides a usable and effective interface for supporting surveillance operators in detecting suspicious behavior.

2.16.1 Demographics of Participants

2.16.2 Questionnaire/Interviews Results

2.17 Discussion

Provide an interpretation of results you presented in section 5. Were the results obtained from the "system evaluation" good or not? Why? How can we improve the process, the design, or the implementation?

CHAPTER.8

Conclusions and Future Work

3 Conclusions and Future Work

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APPENDIX

8. Appendix A :Interviews

Table 8:Interview 1's Results - Release 1

Interview 1	
Question	**Answer / Result**
1. Name & Job Title	Laila – Security Supervisor / CCTV Monitor
2. Years of Experience	14 Years
3. Work Environment	Security / Exhibition
4. Average Number of Cameras Monitored Per Day	12 to 16 cameras per day
5. Type of Surveillance System Used	Traditional System
6. How would you describe your overall experience with the current surveillance systems?	Exhausting; relies heavily on human focus, intuition, and speed; requires constant concentration.
7. What is the most challenging part of your job when it comes to monitoring visitors?	Predicting behavior and distinguishing between normal and abnormal movements.
8. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage?	Never encountered; focus is purely on live monitoring
9. Do you believe that relying solely on human observation makes it difficult to detect small movements in crowds?	Yes, crowds cause distractions; a technical assistant (system) is necessary.
10. How difficult is it to track a person of interest when they move between different camera views?	Tiring to monitor multiple cameras, especially if unorganized.

11. What factors would make you trust and rely on an AI-based surveillance system?	Accuracy; not every human can maintain attention and recognize distractions constantly.
12. During peak hours, how manageable is it for you to monitor all camera screens?	Cannot watch all screens at once; some areas might be missed.
13. When a suspicious situation is identified, what are the usual steps and where do delays occur?	Team communication; delays occur if the receiver is not responsive.
14. What type of alert would be most effective for you during work?	Sound notification combined with a visual alert.
15.How important is it for you that the surveillance system operates locally without cloud transmissio	Prefers local storage (with visual cues) over cloud for security.

Table 9:Interview 2's Results - Release 1

Interview 2	
Question	**Answer / Result**
1. Name & Job Title	Abdulelah Qassem Al-Faifi – Perfume Salesperson
2. Years of Experience	6 Months
3. Work Environment	Exhibition
4. Average Number of Cameras Monitored Per Day	6 cameras
5. Type of Surveillance System Used	Traditional System
6. How would you describe your overall experience with the current surveillance systems?	Primitive; depends fully on human monitoring without advanced technology.
7. What is the most challenging part of your job when it comes to monitoring visitors?	High crowd density; difficult to focus and follow everyone simultaneously.
8. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage?	Yes, it happened more than once (detected after reviewing recordings).

9. Do you believe that relying solely on human observation makes it difficult to detect small movements in crowds?	Yes, human attention might miss fine details consistently.
10. How difficult is it to track a person of interest when they move between different camera views?	Much harder as the number of cameras and customers increases.
11. What factors would make you trust and rely on an AI-based surveillance system?	When the system is precise and accurate in detecting suspicious behaviors.
12. During peak hours, how manageable is it for you to monitor all camera screens?	Very difficult; hard to maintain focus for a long period.
13. When a suspicious situation is identified, what are the usual steps and where do delays occur?	Alerting the customer indirectly; delays occur when the person is at a distance.
14. What type of alert would be most effective for you during work?	Visual alert on the dashboard.
15.How important is it for you that the surveillance system operates locally without cloud transmissio	Very useful for small items to help make faster decisions.

Table 10:Interview 3's Results - Release 1

Interview 3	
Question	**Answer / Result**
1. Name & Job Title	Sultan – Self-Service Security Surveillance Supervisor
2. Years of Experience	18 Years
3. Work Environment	Security / Exhibition
4. Average Number of Cameras Monitored Per Day	7 cameras
5. Type of Surveillance System Used	Traditional System
6. How would you describe your overall experience with the current surveillance systems?	Requires high concentration; reviewing recorded footage is very exhausting.
7. What is the most challenging part of your job when it comes to monitoring visitors?	Monitoring during busy exhibition hours.
8. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage?	Yes, suspicious behavior was detected only after re-watching recorded footage.
9. Do you believe that relying solely on human observation makes it difficult to detect small movements in crowds?	Yes, must re-watch several times to detect small or slow movements.

10. How difficult is it to track a person of interest when they move between different camera views?	Easy due to the small area, but the challenge is monitoring blind spots.
11. What factors would make you trust and rely on an AI-based surveillance system?	Feels confident but believes there should always be a human backup.
12. During peak hours, how manageable is it for you to monitor all camera screens?	Monitored daily; takes an entire day to review recordings properly.
13. When a suspicious situation is identified, what are the usual steps and where do delays occur?	Reviewing recordings; delays occur due to camera blind spots.
14. What type of alert would be most effective for you during work?	A mix of visual dashboard alerts and mobile notifications.
15.How important is it for you that the surveillance system operates locally without cloud transmission	Yes, it would be helpful because it makes identifying events easier.

Table 11:Interview 4's Results - Release 1

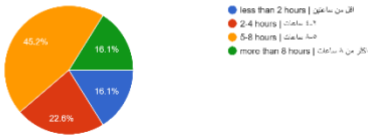
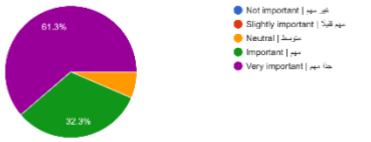
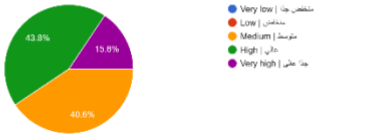
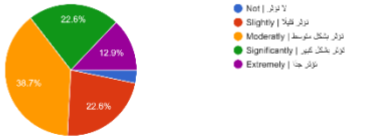
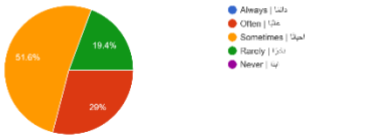
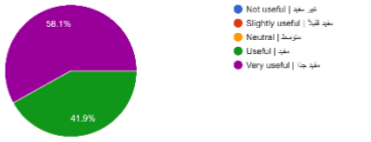
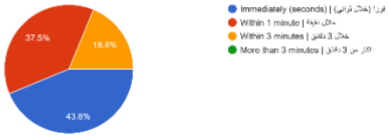
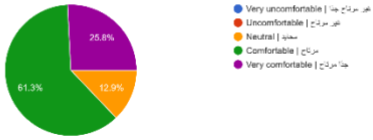
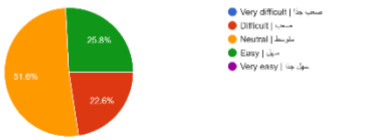
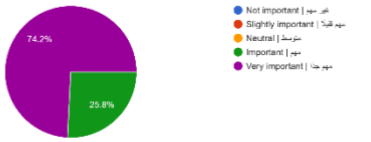
Interview 4	
Question	**Answer / Result**
1. Name & Job Title	Eijan Al-Mandil – Staff
2. Years of Experience	9 Years
3. Work Environment	Security / Exhibition
4. Average Number of Cameras Monitored Per Day	5 cameras
5. Type of Surveillance System Used	Traditional System
6. How would you describe your overall experience with the current surveillance systems?	Easy but not engaging or enjoyable.
7. What is the most challenging part of your job when it comes to monitoring visitors?	Difficult camera angles.
8. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage?	No, but equipment failures occur where the source of the fault is unknown.
9. Do you believe that relying solely on human observation makes it difficult to detect small movements in crowds?	Hard to detect in unclear areas or when there is no audio.
10. How difficult is it to track a person of interest when they move between different camera views?	Space is small, so no major difficulty except for blind spots.
11. What factors would make you trust and rely on an AI-based surveillance system?	Trusts it for everything; useful for providing detailed reports on all activities.

12. During peak hours, how manageable is it for you to monitor all camera screens?	Becoming easier now as it can be monitored via mobile.
13. When a suspicious situation is identified, what are the usual steps and where do delays occur?	Delays occur due to internet issues or crowded spaces.
14. What type of alert would be most effective for you during work?	Both sound and visual alerts together for better clarity.
15.How important is it for you that the surveillance system operates locally without cloud transmissio	Prefers local storage over cloud for security.

9. Appendix B: Questionnaires

Questionnaires Questions:

1. How many hours per shift do you spend monitoring cameras?
2. During monitoring, how high is your mental workload or focus level required?
3. How often do you feel that important events might be missed during live monitoring?
4. How quickly do you usually need to react when suspicious behavior is noticed?
5. How clear and easy to use is your current surveillance system interface?
6. How important is it for you to receive alerts with visual evidence (snapshot or short clip)?
7. How much do false alarms affect your work efficiency?
8. How useful would it be if the system prioritized the riskiest behavior automatically?
9. How comfortable are you with AI assisting you in monitoring tasks?
10. How important is system reliability (working without crashes or delays) during monitoring?
11. What features would help you perform your security tasks better?

Q	Result	Q	Result
1	1- How many hours per shift do you spend monitoring cameras? كم ساعة تقضي في مراقبة الكاميرات خلال الورديه الواحدة ؟ 31 responses  	6	6- How important is it for you to receive alerts with visual evidence (snapshot or short clip)? ما مدى اهمية استلام تنبيه يحتوي على دليل مرئي (صورة أو مقطع قصير)؟ 31 responses 
2	2- During monitoring, how high is your mental workload or focus level required? اثناء المراقبة ، ما مستوى التركيز أو الجهد الذهني المطلوب منك ؟ 32 responses  	7	7- How much do false alarms affect your work efficiency? إلى أي مدى تؤثر الإنذارات الخاطئة على كفاءة عملك؟ 31 responses 
3	3- How often do you feel that important events might be missed during live monitoring? كم مرة تشعر انه قد يتم تفويت احداث مهمة اثناء المراقبة المباشرة ؟ 31 responses  	8	8- How useful would it be if the system prioritized the most risky behavior automatically? ما مدى فائدة ان يقوم النظام بإعطاء أولوية تلقائيا لأخطر السلوكيات؟ 31 responses 
4	4- How quickly do you usually need to react when suspicious behavior is noticed? ما السرعة المطلوبة عادة للاستجابه عند ملاحظة سلوك مشبوه ؟ 32 responses  	9	9- How comfortable are you with AI assisting you in monitoring tasks? ما مدى راحتك في استخدام نظام ذكاء اصطناعي لمساعدتك في المراقبة؟ 31 responses 
5	5- How clear and easy to use is your current surveillance system interface? ما مدى وضوح وسهولة استخدام واجهة نظام المراقبة الحالي؟ 31 responses  	10	10- How important is system reliability (working without crashes or delays) during monitoring? ما مدى اهمية موثوقية النظام (العمل بدون تعليق أو تأخير) اثناء المراقبة؟ 31 responses 

11	**11- What features would help you perform your security tasks better?** **ما الميزات التي قد تساعدك على أداء مهامك الأمنية بشكل أفضل؟** 14 responses Summarize responses -Risk level prioritizations -mukti camera suspect tracking -snapshot evidence with alert ان يكون هناك تنبية صوتي وأشعار عند حدوث شي مشبوه - Clear location of suspensious activity - Fewer false alarms وجود سجل أحداث يمكن الرجوع له التنبيه خلال وقت قصير جدا - Silent alarms on dashboard -Record automatic snapshots for future use and documentation. -System that works fully offline -behavior based alerts (not just motion) وجود منبة -A "false alarm feedback" button so i can train the system يكون هناك نظام مساعد متطور يساعد الفرد فيه التنبيه عند وجود مشتبه وايضا يتتبع المشتبه به لان هناك صعوبه بتتبع جميع الشاشات والكاميرات ١-تحديد موقع الحدث ٢-واجهة سهلة وواضحة -Color coded risk level to help me react faster -One notification instead of multiple for the same behavior يكون هناك مساعد ويتتبع تحركات المشتبة به ان يعمل بدون انترنت، امكانية إرجاع الفيديو

Form's link: https://forms.gle/sD988Lv4VBejDv5aA

10. Appendix C: GitHub

GitHub link: https://github.com/lanaabdullah1/2025-GP1-3.git

11. Appendix D: Jira

Jira link: https://joudkja1.atlassian.net/jira/software/projects/DK2025/boards/3